

DARK MATTER AND DARK ENERGY

THE LUMINOUS MATERIAL OF THE UNIVERSE IS A TINY PART OF ITS OVERALL COMPOSITION—IT ALSO CONTAINS LARGE AMOUNTS OF INVISIBLE MASS KNOWN AS DARK MATTER, AND ANOTHER MYSTERIOUS SUBSTANCE CALLED DARK ENERGY.

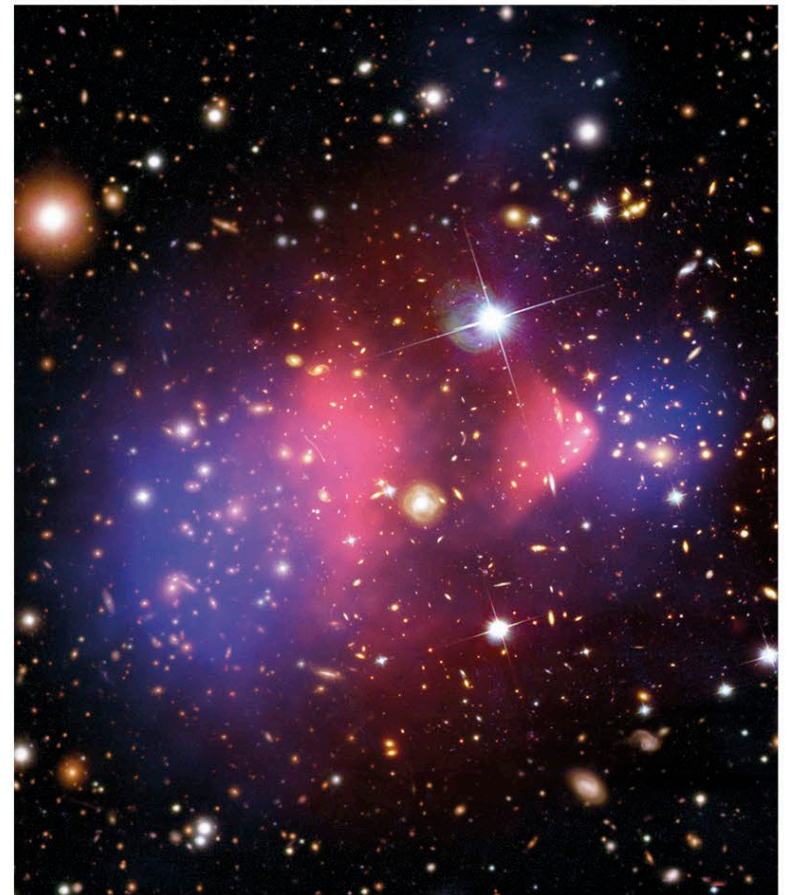
Since the 1930s astronomers have suspected the existence of dark matter. Such material is not just dark, but completely immune to interactions with light, and only makes its influence felt through the force of its gravity. More recently, cosmologists have found that cosmic expansion (see pp.70–71) is accelerated by an effect called dark energy, which seems to counteract gravity.

The nature of dark matter

The first evidence for dark matter came from two sources: the way that galaxies orbit inside galaxy clusters and the speed at which stars orbit in the Milky Way. Both suggest the Universe contains about five times more dark than luminous matter. Some of this may be accounted for by compact, faint objects such as dead stars and stray planets, but most of it probably consists of unknown subatomic particles—tiny objects that interact with normal matter only through gravity.

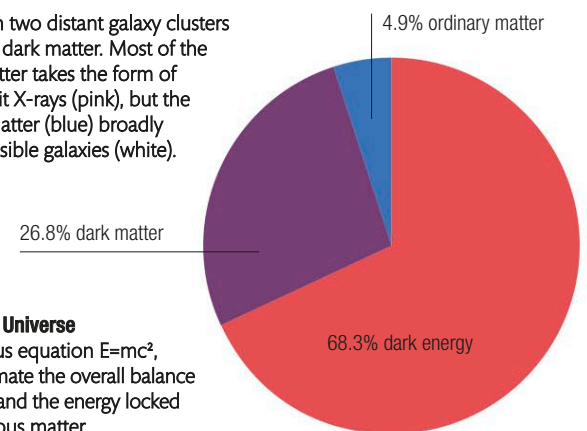
What is dark energy?

In the 1990s, cosmologists discovered that cosmic expansion, which began in the Big Bang (see pp.14–15) has sped up over time, rather than slowing down as we might expect due to the gravity of all the dark and luminous matter within the Universe. The substance that drives this expansion is known as dark energy, but it's still poorly understood—it may be a “cosmological constant” (a uniform property of spacetime itself), or a “quintessence” (a localized force that can vary from place to place). Which theory is correct could have an important effect on the fate of the Universe.



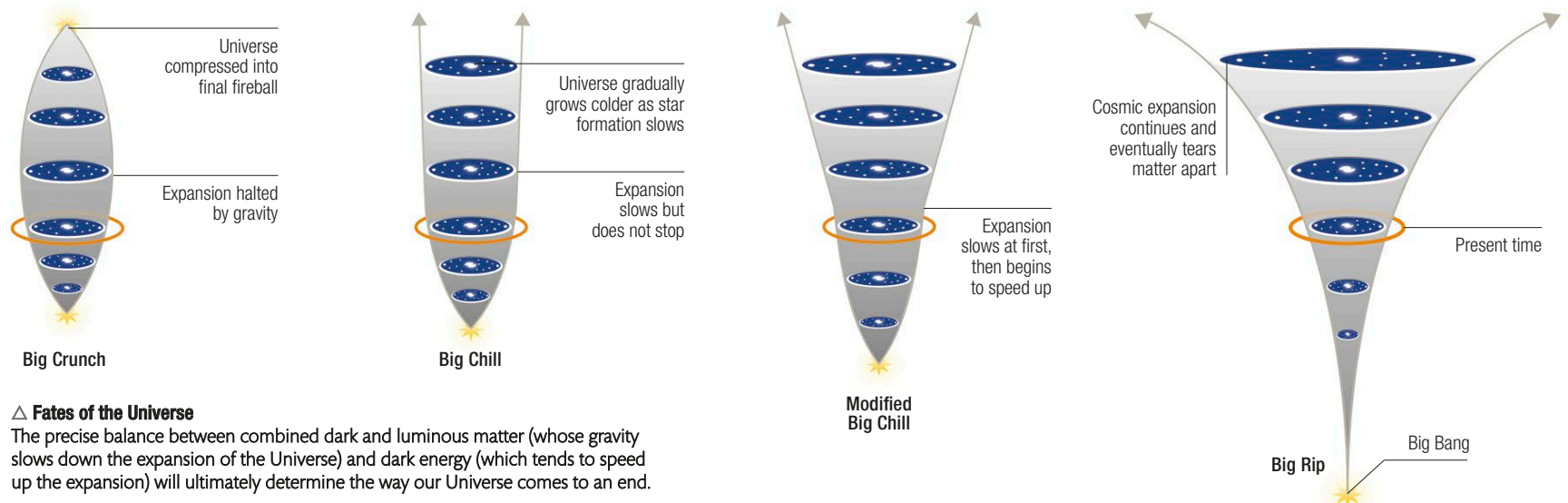
△ Bullet cluster

This collision between two distant galaxy clusters reveals the motion of dark matter. Most of the cluster's luminous matter takes the form of clouds of gas that emit X-rays (pink), but the distribution of dark matter (blue) broadly matches that of the visible galaxies (white).



▷ Composition of the Universe

Using Einstein's famous equation $E=mc^2$, cosmologists can estimate the overall balance between dark energy and the energy locked up in dark and luminous matter.



△ Fates of the Universe

The precise balance between combined dark and luminous matter (whose gravity slows down the expansion of the Universe) and dark energy (which tends to speed up the expansion) will ultimately determine the way our Universe comes to an end.